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Real-Time Fluid Contact Identification Using Latest Intelligent Dynamic Reservoir Evaluation Technology Extended Reach Horizontal Well in Carbonate Reservoir, a Case Study in a Carbonate Reservoir, Offshore Abu Dhabi

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Abstract

Production wells in mature offshore fields encounter many challenges. In this study the main consideration is avoidance of early water breakthrough driven by injection/production. Encountering water complicates effective completion and production operations. Real-time understanding of geology and fluids, particularly water saturation variance, is required to optimize well placement. This case study is from a recent well in a mature field offshore UAE. Reservoir fluid identification solution utilizing the latest technology was recognized as the most effective method to confront the challenges in this well.

The well target was planned for approximately 12,100 feet, targeting a specific carbonate layer in an 8.5″ open hole horizontal section. Precise geosteering required that is a combination of logging while drilling technologies: Triple Combo; and a borehole Imaging Azimuthal Density tool (for detailed formation evaluation). The focus on real-time fluid identification supported any increasing proximity to the conductive water zone, highlighting areas that could cause water flooding during production. During well placement and steering operations, it has been provided for this well through the whole operation from the planning stage; through the execution stage (Realtime Modelling); and post well analysis as validation of steering actions post TD.

During drilling the 8.5″ hole section several geological and drilling challenges were encountered. Pro-active maneuvering of the trajectory to land in the reservoir's optimum zone was required in the early interval due to setting of the previous section's casing shoe in the dense layer above the target reservoir. The top boundary above the target formation was mapped allowing continual interpretation of formation dip, which was in the range of 1.2 – 1.7 deg up towards the toe. Simultaneously, the bottom boundary of the target reservoir was continuously mapped to TD 7 – 9 ft below trajectory and this is where the main challenge was identified with the identification of a conductive water zone with increased proximity to the wellbore.

Effective action by utilizing double latch to the ORA tool in tough logging condition had enabled the oil water contact to be identified and acquiring the optimum hydrocarbon bearing zone and evaluating the toe of the horizontal hole to verify the water existence. Geosteering services and quick decision making had

facilitated the rapid and critical decision to obtain and deliver the well objective, to allow the possibility of water leg to be seen in the well and all over the field. To continue drilling and completion without identifying nearby water movement in this section would result in significant water production that will enhance the injection mechanism. This would be possible using conventional electromagnetic tools. In later stage, production of this well had provide promising results, securing at least 4 years of maximizing well life and expanding the hydrocarbon reserve column. This practice shall set a new standard for future appraisal and mature reservoir wells.

Introduction

Field Background, UAE Offshore

The developed offshore field of this case study, known as Field-F, has been developed for oil production since the 2019's. The main reservoir is extracted from the Upper Jurassic era formation, focusing on the lower part that consists of several layer reservoir units, in which the lithology mainly consists of alternation of dolomite and limestones. The study of the depositional system discloses a gentle carbonate ramp with low angle as a repercussion of an abrupt transition from an open sea depositional system to a temporarily confined depositional system -- resulting from the insulation by a belt of an intra-shelf basin, from open sea influences. The main diagenetic processes are related to dolomitization with a minor impact due to incipient cement precipitation, hence layers of reservoir units within this lower part formation had several indications of anhydrite cementation. Numerous horizontal barriers created by the anhydritic beds have led to the presence of several oil-water contacts and variable, but generally poor, natural aquifer pressure support. In addition, a few years of productivity/injectivity had also mis-impacted the increase of water saturations in each reservoir unit. Hence, the main challenge of production decline in this Field-F comes from the natural water leg which is not seen in most of the drilled wells.

The intergranular primary porosity preserved in limestone deposits, associated with vugs and molds resulting from bioclasts leaching. Fracture porosity is preferentially observed in dolomites, but some exist in limestones. This case study's Target Layer F1- formation are considered as limestone/dolomite units with high porosity (> 16%) and poorly fractured. From the regional study, in un-faulted zones only open NE-SW fractures develop in less porous-thin layers.

Horizontal well design had been implemented since the 2019's in this Field-F to target bypassed reservoirs and increase reserves from its early maturing and negligible water-cut oil field. The case study that shall be elaborated in the next section will discuss thoroughly the advancement in well data acquisition solutions by utilizing the latest and fit-for-purpose technology to effectively and economically achieve maximum production and injection.

LWD/WL for Well Placement Solutions

Well placement solutions in the oil and gas market have been growing rapidly in the last decade. In the recent LWD (logging-while-drilling) technologies deployment in reservoir well placement and in the integration of Near-Field and Far-Field measurements. Various field challenges can now be handled promptly in discerning nature through comprehensive analysis during realtime drilling. This case study will focus on the implementation of reservoir evaluation technology with WL/ORA tool (Fig.1) drill pipe conveyed deployment to support and improve the current oil recovery technique in Field-F.

Figure 1—ORA Intelligent Wireline Formation Testing Platform consists of four pilar for dynamic reservoir evaluation

Case Study

Pre-job Preparation

Well placement solution starts with the preparation of analyzing offset wells, to gain detailed understanding of the optimum zone's properties within the target layer. Target well was planned to be at the North-East part of the field (Fig.2), in which only zero well was not identified with the oil water contact (Fig.3). Target-Well was targeting layer F1-formation for its horizontal placement derived from a slanted deviated profile well. The properties analyzed from offset well it was clearly seen that the optimum zone shall be in the middle part, approximately ~ 2 – 4 ft.TVT from top boundary. The entire thickness of F1-layer was known to be ~ 6 – 7 ft.TVT. As per the log properties, it was identified that the optimum zone will be revolved around the resistivity of 3 – 7 Ohm, with rapid decrease towards the base of F1-layer and was confirmed as the resistivity properties of water. This low resistivity was not expected due to years of oil productivity and zero natural water flooded in these layers all over the field. From geology perspective, the structural was expected to be down dipping based on the geophysical evaluation. However, high uncertainty was still present from the field's experience with seismic data available. The Well Plan had been designed to target the horizontal placement in 8.5″ hole section, with lateral footage of ~12,100 ft. The start of this lateral section was continuing the end of 8.5" total depth, which placed the well still in the reservoir layer the target F1-formation with a smooth sloped an angle.

Figure 2—zero well was not identified the oil water contact

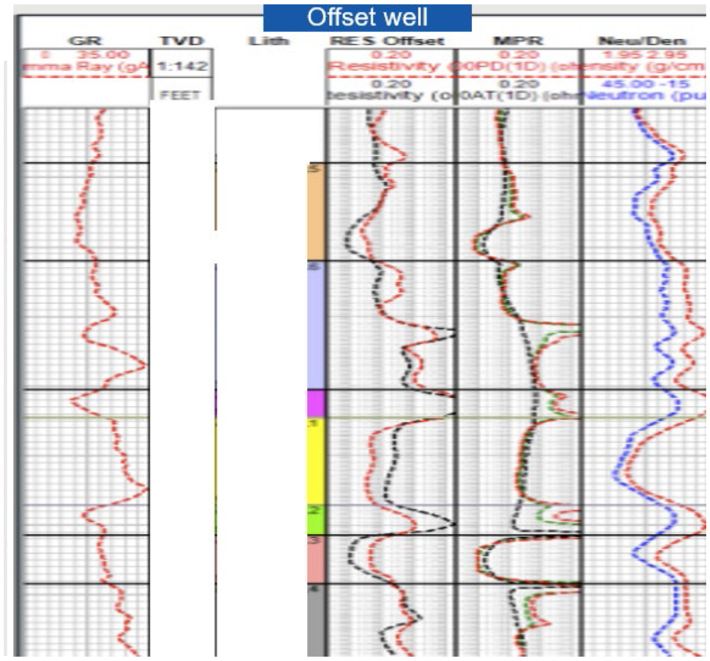

Figure 3—Offset Well Analysis

As the well placement's challenges had been identified, the next phase was to select the proper geosteering sensor to overcome and strategize the entire steering. Fluid identification capabilities were found to be the fit solution, with ORA as the main sensors. This selection was made to achieve the objectives of allocating water breakthrough or water-flooding from layer below, whilst simultaneously mapping the roof of reservoir to maintain precise well placement in the upper part of the layer. Resistivity contrast was seen clearly from the offset log, portraying high-level confidence for resistivity to distinguish between risk and target zone desired. Various reservoir fluid scenarios were generated in this pre-well modelling stage to simulate possible real-time outcomes from all aspects of the challenges.

The main output of pre-well modelling stage was the agreement of geosteering strategy based on the deep resistivity simulated response. The software utilized offset well logs and surface files as the main input, then processed through a variogram algorithm which generated a resistivity-pseudo log along the well plan. Subsequently, pre-well modelling was processed, testing diversity of tool settings and processing parameters. Through comprehensive analysis, the effective settings of spacings and frequencies were determined to achieve geo mapping. The focus was to achieve vivid resolution of to resolve the resistivity contrast of mapping above and below the well trajectory. Furthermore, ORA tool had been agreed to be activated in real-time mode to track precisely the fluid movement surround the reservoir.

The outcome of pre-well modelling scenario present the potential ability to map up to 2Mhz & 400kHz electromagnetic wave, long spaced and short spaced receivers resistivity layers (Fig.4), hence able to highlights the resistivity profile surrounding F1-target layer: higher value on the middle part of target representing optimum zone; lower value in the base part of target representing water properties; and much higher value above the target representing the dense layer properties.

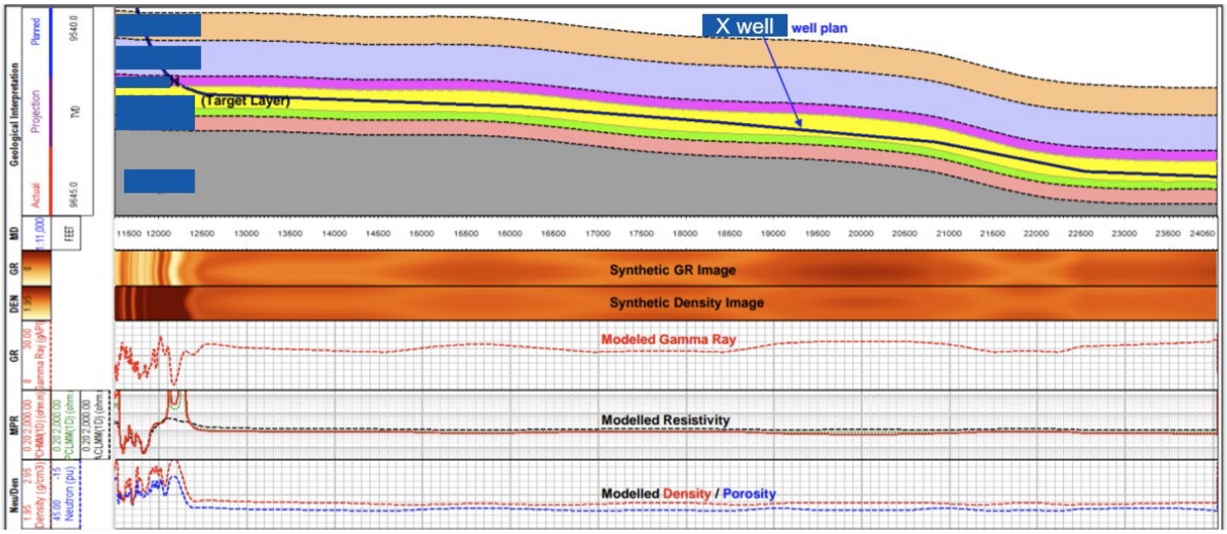

Figure 4

Hence, the geosteering strategy had been set in this pre-job preparation stage. BHA (bottom-hole-assembly) configuration was effectively finalized to achieve well placement and formation evaluation objectives (Fig.5): ORA tool; Triple Combo. A steering decision matrix, based on target and risk identification, had been made and shall be followed in flowchart format to support safety operational decision making.

Figure 5—Final BHA configuration deployed for Case Study - Target Well

The well trajectory (see Fig. 6) posed significant challenges for the deployment of a wireline formation testing tool, primarily due to its extended reach and high deviation. To effectively address these challenges, a drill pipe conveyance method was employed. In this approach, the tool was conveyed to the casing shoe using drill pipe, after which a wireline cable was run and securely attached to the tool. This setup enabled the establishment of power and communication for tool operation. A Cable Side Entry Sub (CSES) (see Fig. 7) was utilized to allow the wireline cable to pass through the drill pipe while maintaining connectivity between the tool and the drill pipe. This configuration ensured the safe deployment of the tool up to a deviation of 45 degrees. However, beyond this deviation, the risk of cable damage between the drill pipe and casing significantly increased.

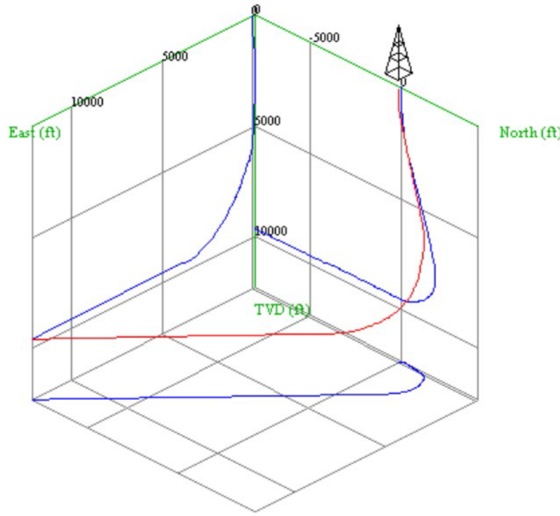

Figure 6—Nearly 12000 ft horizontal OH section where casing shoe at 90 deg deviation

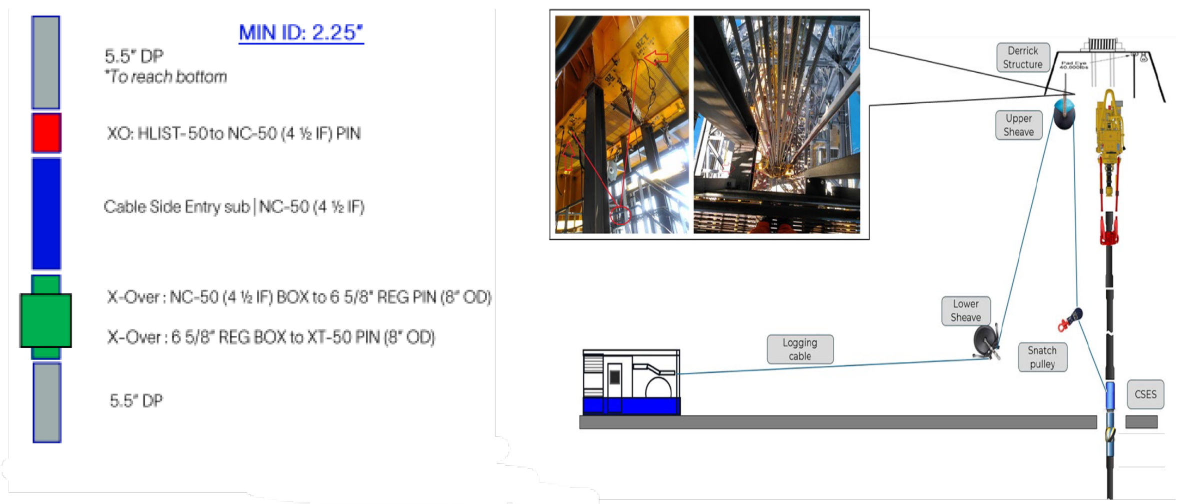

Figure 7—Cable Side Entry Sub (CSES) connected to drill pipe to access cable to connect cable with wireline tool

In this specific well, the deviation reached 45 degrees at a measured depth of approximately 9,100 ft, leaving over 3,000 ft of the open-hole interval unevaluated if conventional deployment methods were used. To overcome this limitation, a double-latching deployment strategy was introduced. This approach enabled the tool to evaluate the initial 4,000 ft of the open-hole section during the first run. The tool was then tripped back to the casing shoe, where the CSES was disconnected. Subsequently, the tool was re-run into the wellbore to the previously evaluated 4,000 ft depth in the open hole. At this point, the CSES was reconnected, allowing for safe deployment of the tool to total depth (TD). This method facilitated the acquisition of critical formation data across the entire open-hole interval, ensuring comprehensive reservoir evaluation.

One of the most critical operational challenges encountered in this well was the risk of the tool becoming stuck during stationary measurements, a known concern inherent to wireline formation testing operations. This risk was further amplified in an extended-reach well, where the high deviation and long horizontal section could significantly complicate retrieval efforts, potentially leading to prolonged and costly fishing operations. Historical experience in similar wells underscored the importance of addressing this issue proactively to ensure operational success and mitigate non-productive time (NPT).

To effectively manage this risk, a specialized ORA DTT configuration was employed. This setup provided a robust safe pull capacity of 100,000 lbf, ensuring sufficient overpull capability to free the tool in the event of potential sticking. This configuration was specifically designed to address the mechanical challenges associated with extended-reach wells, offering the operational flexibility required to handle unforeseen downhole conditions.

Moreover, the ORA tool itself featured a dual flowline system equipped with two large inlet and focused flow capabilities, two pump and fluid insitu scanner with lab accuracy metrology to identify fluid properties (see Fig. 8). This significantly optimized the efficiency of fluid identification, thereby reducing the stationary time required for measurements. By minimizing the duration that the tool remained stationary, the likelihood of sticking was dramatically decreased. This dual flowline system also enhanced the quality of collected data, as focused flow enabled more accurate reservoir characterization by differentiating formation fluid from drilling mud filtrate during acquisition.

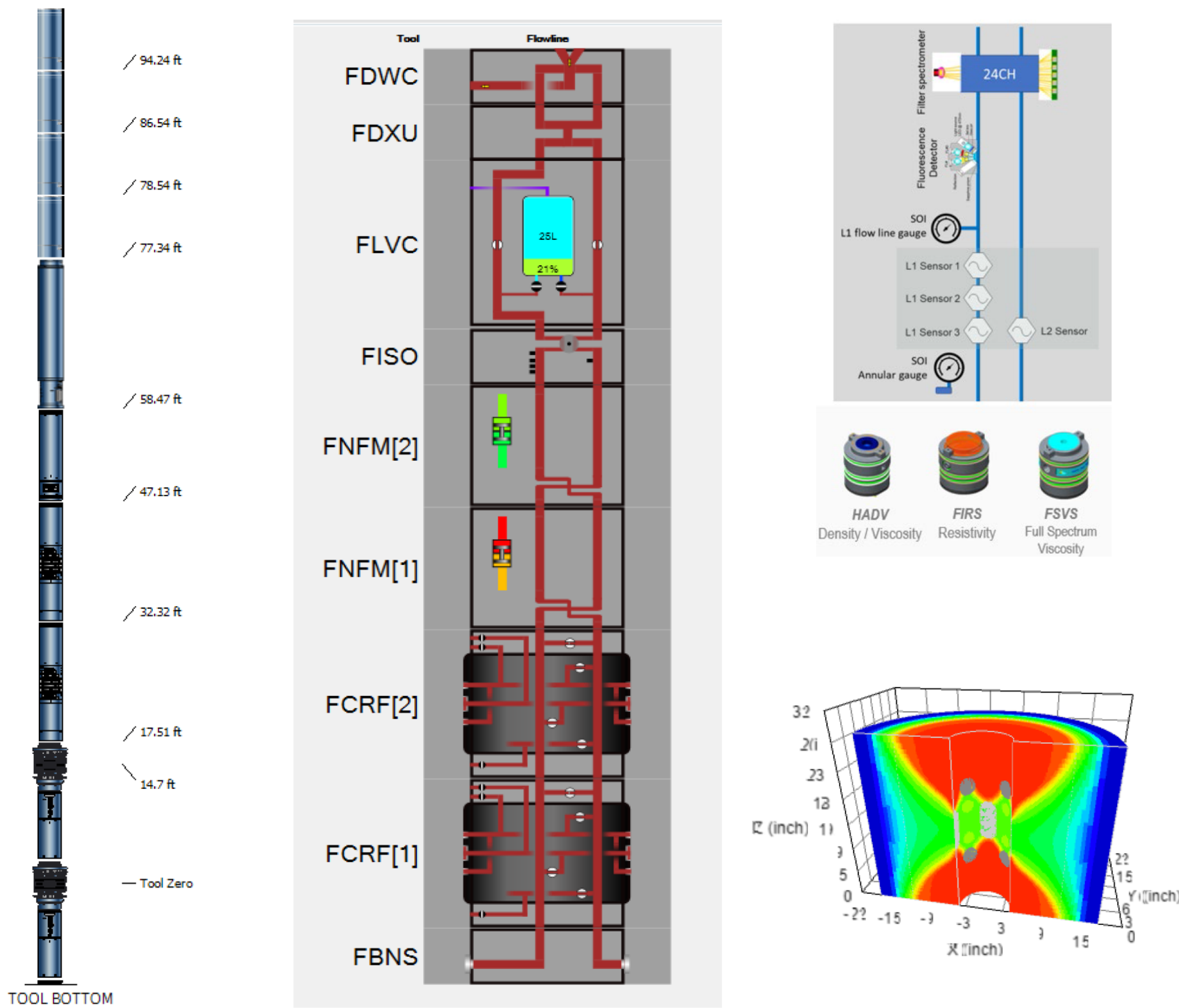

Figure 8—ORA Intelligent Wireline Formation Testing Tool configured with two focus inlets, two flow pumps and fluid insitu scanner.

The combination of the high-capacity DTT setup and the ORA tool's advanced flowline system not only mitigated the risk of sticking but also provided the drilling team with the confidence needed to safely deploy the tool in this challenging environment. These technological advancements ensured the successful acquisition of critical reservoir data while minimizing operational risks and associated costs, ultimately contributing to the overall efficiency and reliability of the wireline formation testing operation in this extended-reach horizontal well.

Job Execution

Following the rig-up of the toolstring, a thorough function test was conducted at the surface to ensure optimal performance before deployment. The toolstring was then conveyed to the casing shoe using drill pipe. Once the tool reached the casing shoe, a second function test was performed to verify its operational integrity under downhole conditions. After confirming functionality, the Cable Side Entry Sub (CSES) was connected, and the wireline survey commenced, progressing from the casing shoe (CS) toward the total depth (TD). The operational plan included a total of fifteen stationary measurement stations, with three of these dedicated to fluid identification in order to establish the oil-water contact (OWC).

During the operation, six stationary stations were successfully completed, recording pressure data at discrete intervals within the reservoir. One of these stations was also utilized for fluid identification, which confirmed hydrocarbons as the primary reservoir fluid. Despite a relatively low mobility of 2.5 millidarcies per centipoise (md/cp), the tool demonstrated impressive efficiency, pumping a total of 240 liters of formation fluid within a 2-hour pumping period. This achievement highlights the tool's advanced capability to handle low-mobility formations, ensuring reliable fluid sampling without compromising operational timelines. (See Fig.8a)

Figure 8a—First Fluid Identification station indicating formation oil

At this stage of the operation, the tool was pulled back to the casing shoe, where the Cable Side Entry Sub (CSES) was disconnected. The tool was then run in hole (RIH) into the open hole (OH) without the cable attached, reaching the depth of the last completed station. Once at this depth, the CSES and wireline cable were reconnected, enabling communication and power for the subsequent measurements.

The remaining nine planned stationary measurement stations were successfully executed, including two additional fluid identification tests. The first fluid identification confirmed hydrocarbons as the dominant formation fluid, with only 5-10% formation water present (See Fig. 9).

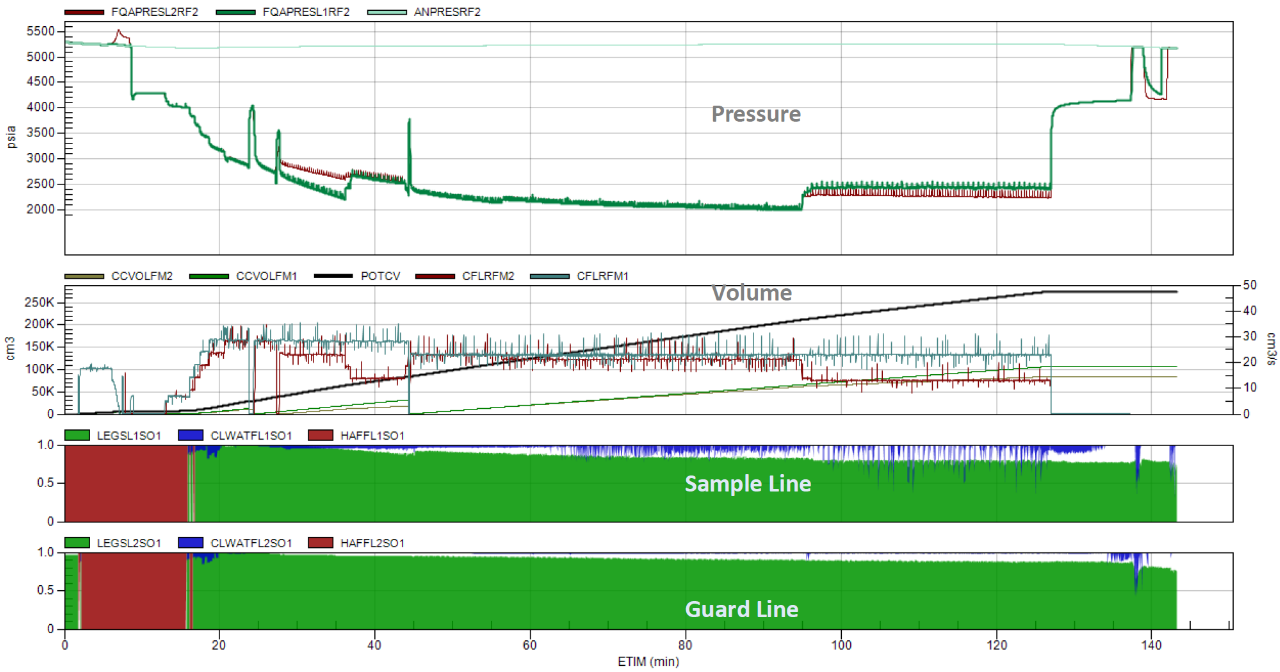

Figure 9—Second Fluid Identification station indicating formation oil with 5-10% formation water

The third fluid identification station predominantly indicated the presence of water as the formation fluid. After approximately 20 minutes of pumping at this station, the guard line lost its seal. At this point, over 80% of the flowing fluid was confirmed to be water. To continue the operation, pumping was carried out using the sampling line. Despite pumping only one flowline, more than 200 liters of fluid were successfully pumped, which was analyzed and identified as formation water with approximately 5% oil-based mud filtrate. (see fig. 10)

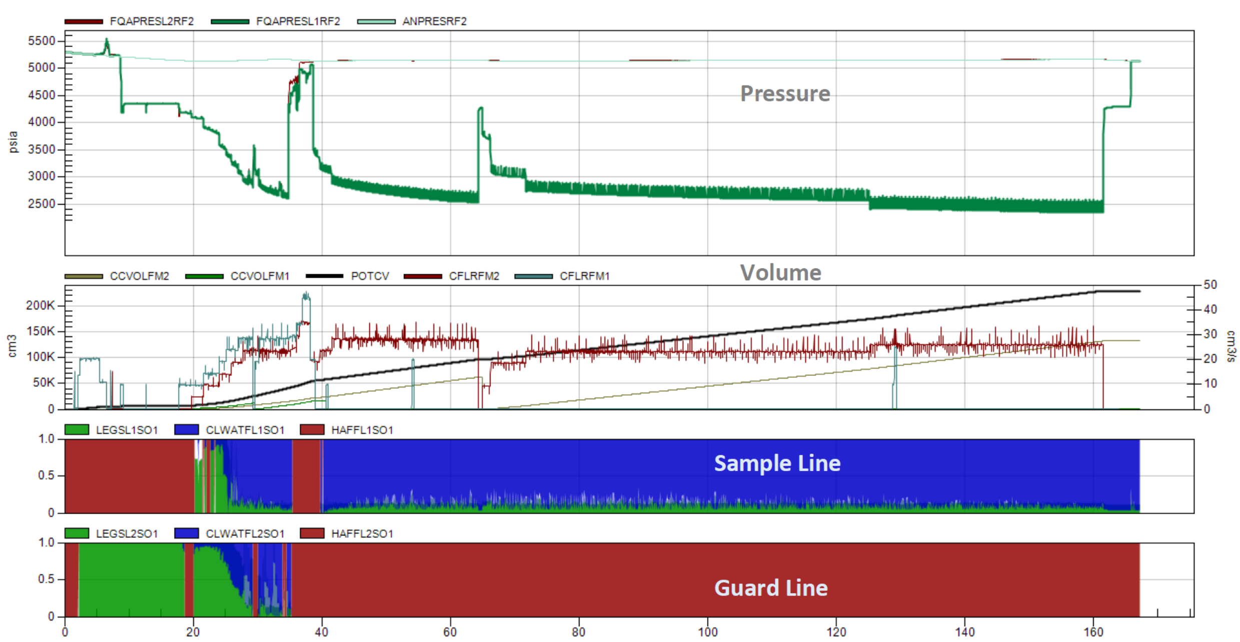

Figure 10—Third Fluid Identification station indicating formation water with 5% oil based mud filtrate

Following the observation of water at the fluid identification station, one additional pressure measurement was performed. This final step successfully fulfilled the primary job objective by confirming the oil-water contact (OWC). The operation was concluded with all objectives met and a complete dataset capturing the reservoir's fluid distribution and pressure characteristics. (See fig 11)

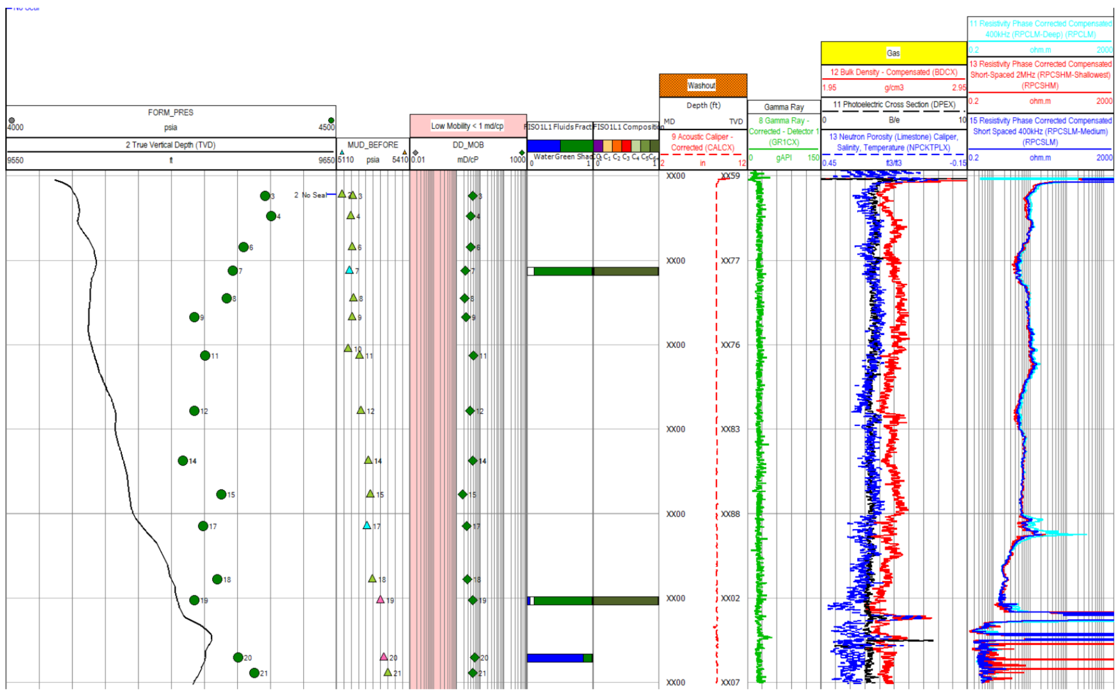

Figure 11—ORA Intelligent Wireline Formation Testing Pressure and Fluid Identification depth view showing pressure and fluid identification result.

Conclusion

Incorporating ORA testing into the operational workflow was essential for validating geosteering decisions throughout the drilling process. The acquisition of real-time fluid data via the ORA tool enabled the team to verify hydrocarbon presence within the target zone and identify any signs of water encroachment. This critical information supported the optimization of well placement strategies and contributed to securing the well's long-term productivity. The ORA tool mapped reservoir fluid movement, boosting confidence in meeting well objectives and setting a standard for future Field-F operations.

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